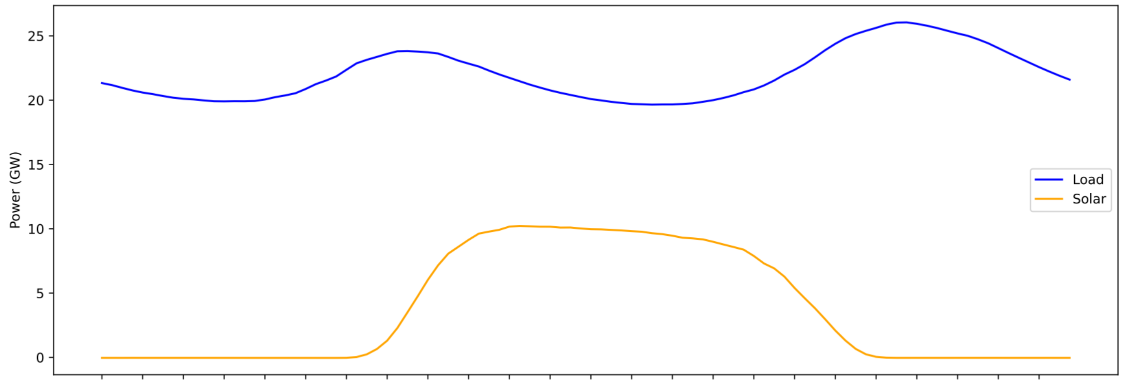
Convex Optimization

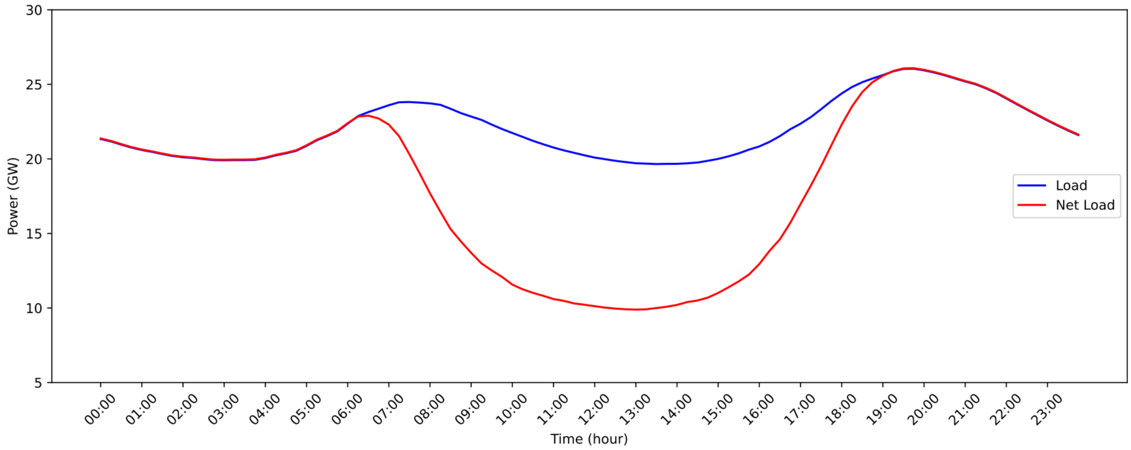
Project 2

Spring 1403

Due date: 15th of Ordibehesht

1. *The duck curve and storage.* The plot below shows the total daily electricity load (or demand) and the total solar generation for California, averaged over March 2021, in GW (gigawatts). The plot below that shows the load and also the net load, the load minus the solar generation, which must be supplied by other generation sources. If you squint your eyes, the load and net load curves look like a duck, hence the name 'duck curve'.





The steep rise in net load around 5-7pm can be a challenge for generators, which work better and more efficiently when generating constant or smoothly varying power. Many generators have a ramp rate limit, *i.e.*, there is a limit on how fast they can change their power over time. In this exercise we consider using storage (batteries) to reduce the cost of supplying the net load and to reduce the maximum ramp rate.

The work with quantities in 15 minute (0.25 hour) intervals over one day, which is 96 periods, with $t = 1$ corresponding to the interval $0:00 - 0:15$ and $t = 96$ corresponding to the interval $23:45 - 24:00$. The storage device has energy q_t, $t = 1, ..., 96$ with $0 \leq q_t \leq Q$, where $Q = 15GWh$ (gigawatt-hour) is the capacity of the storage. In period t we charge the battery with power c_t (in GW), with negative values of c_t denoting discharge. The maximum charge and discharge power is C, *i.e.*, $\|c_t\| \leq C$, $t = 1, ..., 96$, with $C = 3.73GW$. This means that the battery can be charged from empty to full, or discharged from full to empty, in 4 hours. The battery energy satisfies

$$q_{t+1} = q_t + 0.25c_t,\ t = 1, ..., 95,\ q_1 = q_{96} + 0.25c_{96}.$$

(The 0.25 factor gives the energy in GWh for a constant power over a 15 minute period, and the last constraint requires that the storage device starts and ends with the same energy.)

We denote the load as ℓ_t, the solar generation as s_t, and the net load as n_t. These are related as $n_t = \ell_t - s_t$, $t = 1, ..., 96$. The (non-solar) generation is denoted g_t, $t = 1, ..., 96$. It must equal the net load plus the battery charging power, *i.e.*,

$$g_t = n_t + c_t,\ t = 1, ..., 96.$$

The variables are q_t, c_t and g_t, $t = 1, ..., 96$. The data are available in ***duck_curve_data.py***.

The objective is the cost of generation, which is quadratic,

$$J^{cost} = \sum_{t=1}^{96}(ag_t + bg_t^2),$$

with $a = 0.1\ \$10^6/GW$ and $b = 0.005\ \$10^6/(GW)^2$. We also limit the maximum ramp rate, defined as

$$J^{ramp} = 4max\{|g_2 - g_1|, ..., |g_{96} - g_{95}|, |g_1 - g_{96}|\},$$

in GW/h. (The factor 4 converts the change in power to hourly.)

Find the charging, storage, and generation profiles that minimize cost, subject to reducing the maximum ramp rate by a factor of at least two compared to the maximum ramp rate with no storage. Report the cost and maximum ramp rate with and without storage, to two significant digits.

Plot the optimal generation and net load versus time on the same plot. Plot the optimal charging versus time, with two dashed horizontal lines showing the charge limits $\pm C$. Plot the storage energy versus time, with the storage capacity Q shown as a dashed horizontal line.

Remarks. (Not needed to solve the problem.)

- The storage capacity corresponds to about 10% of California residences having a $15kWh$ (kilowatt-hour) battery such as a Tesla Powerwall or equivalent.
- The formula for cost of generation is roughly accurate.
- By the time $15GWh$ of storage could be brought online, there will be even more solar power, further deepening the duck curve.

Good Luck!